

Deepak Mallampati

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Kent, OH (open to Mountain View relocation) - US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer building production ML infrastructure - FastAPI/Docker service packaging, evaluation pipelines, structured logging, and CI/CD for safety-critical model deployments. ~30% post-deploy defect reduction via load simulation and observability. Real-time computer vision experience with YOLOv8 driver-drowsiness detection (~30% inference latency reduction). Sensor-rich ML at the edge (Arduino + walk-forward forecasting). Strong Python + PyTorch + Docker + Jenkins CI/CD + Grafana observability fundamentals. US-based, F-1 OPT, open to sponsorship.

CORE COMPETENCIES

ML Infrastructure • PyTorch • Docker • CI/CD (Jenkins) • Evaluation Pipelines • Computer Vision (YOLOv8) • FastAPI / Flask • Grafana Observability • RAG / Agentic Workflows • Python / SQL

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - USA (Healthcare Technology, AI Consulting, SaaS)

- Built **Retrieval-Augmented Generation (RAG) systems** for clinical knowledge retrieval and healthcare documentation search; recursive semantic chunking + transformer embeddings improved contextual retrieval precision by ~35% and cut irrelevant retrieval >30%; grounded response alignment exceeded 90%.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; raised agent response stability ~25%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement, and support prioritization; improved recall 15-20% on high-risk categories while holding precision >90% via class weighting, stratified sampling, threshold calibration.
- Packaged ML/LLM inference as **FastAPI/Flask** services, containerized with **Docker**, with structured logging and load simulation; reduced post-deploy defects ~30%.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, support ticket logs; raised dataset reliability above 98% and cut downstream model instability ~40%.
- Maintained HIPAA-conscious data governance: de-identification, lineage docs, evaluation audit trails, stakeholder-facing limitation documentation.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India (Oil & Gas / ERP)

- Optimized SQL and **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform; reduced query execution time ~20% and data retrieval latency ~25%.
- Built **CI/CD pipelines with Jenkins** for schema and stored-procedure deployments; cut deployment errors >30% and release cycle ~35-40%.
- Designed performance dashboards via SQL Server DMVs and **Grafana** for long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~25%.
- Tuned database maintenance (index rebuilds, statistics updates, partition-aware indexing); improved reconciliation job runtime ~15%, reduced deadlocks ~15%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Hyderabad, India (Nonprofit / Video Analytics / Applied NLP)

- Built a **transformer-based video summarization & highlight system** over 5,000+ recorded sessions; hierarchical long-context summarization + timestamp-aligned clip extraction reduced manual review time 60-70%.
- Achieved ~85% **alignment** between AI-selected highlights and human-curated key moments.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); cut hallucinated outputs ~30% and raised contextual accuracy >85% in controlled tests.
- Deployed as a **Flask**-based API with a lightweight UI for non-technical staff.

SELECTED PROJECTS

Driver Drowsiness Detection System - Real-Time YOLOv8 Fatigue Monitoring

Replaced a two-stage CNN pipeline with a **unified YOLOv8** detection-and-classification model; reduced inference latency ~**30%**. Sliding-window confidence aggregation to suppress blink-driven false positives (~**25% reduction**), adaptive frame skipping, NMS tuning for stable real-time operation via OpenCV.

IoT + ML Smart Building Indoor Temperature Prediction System

Arduino + DHT11 + DS3231 sensor pipeline with PLX-DAQ logging; time-series forecasting with walk-forward validation across Random Forest, SVR, and Linear Regression. Lag-window feature engineering ($n_{in}=6$); RF achieved ~**20-30% lower MAE** vs linear baseline.

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity search for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces.

Agentic Pixel Character Synthesis & Animation Engine ongoing

Phased generative-AI system for identity-consistent, pose-controlled pixel character synthesis with sprite-sheet export. Stable Diffusion fine-tuning + LoRA for identity, ControlNet + OpenPose/MediaPipe for pose conditioning, latent-space consistency for animation smoothness, and an **LLM-based agent orchestrator** decomposing prompts into generation tasks. Backend: PyTorch, Hugging Face Diffusers, FastAPI, Docker.

Geospatial Road Network Analysis & Community Detection System

Parsed raw OpenStreetMap XML into a weighted **NetworkX** DiGraph; Dijkstra shortest-path, betweenness centrality on a 20k-node subgraph; Louvain community detection and MST extraction for backbone analysis. ~30% runtime reduction via subgraph scoping.

EDUCATION

M.S., Kent State University - Ohio, USA

Aug 2023 - May 2025

WORK AUTHORIZATION

F-1 OPT - authorized to work in the United States; future sponsorship required.

SKILLS

ML Infrastructure: Docker, FastAPI, Flask, Jenkins, CI/CD, MLOps/LLMOps, RESTful APIs, observability/logging, structured logging, load simulation

ML Libraries & Training: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, Diffusers, walk-forward validation, threshold calibration

LLM & GenAI: RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, embeddings, vector search, guardrails

Languages: Python, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

Data & Analytics: Pandas, NumPy, feature engineering, time-series forecasting, NetworkX

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision